

Better face tracking with Singularity + Qpro

A beginner guide for an already rooted Quest Pro

USB CONNECTION REQUIRED

Qpro-Enhanced-FT currently works **only over USB**. Keep a USB data cable connected to the rooted headset while its enhanced eye and tongue tracking runs. In the owner's setup, Virtual Desktop's **Beta** channel also streamed PCVR over USB. The cable stays attached for both.

WHAT THIS GUIDE COVERS

1. Installing **Magisk OverlayFS**, then **Quest Pro Independent Eye Gaze** in Singularity.
2. Installing and running **Qpro-Enhanced-FT** for eye convergence and richer tongue data.
3. A clean test path, safe shutdown, and the known firmware limits.

These are two separate eye-tracking modifications. The Qpro app does not list the Magisk eye module as a requirement, and their combined behavior is not documented. Test each on its own first.

Download what you need

CHOOSE THE CORRECT QPRO DOWNLOAD

On the Qpro GitHub Releases page, expand **Assets** and select the full release package. The automatic **Source code (zip)** is not the runnable app. The Magisk modules are installed inside Singularity.

EXTERNAL DOWNLOAD FOR THIS GUIDE

Qpro-Enhanced-FT release package - download

Choose QproFaceTracking-<version>-poc.zip from Assets; extract the whole folder.

INSTALL THESE IN SINGULARITY

Open **Apps/Modules > Magisk repo**. Install **Magisk OverlayFS** first, then **Quest Pro Independent Eye Gaze**. See page 4 for the full sequence. No separate module ZIP download is needed for this route.

Singularity's Magisk Modules list - reference

The project lists both Magisk OverlayFS and Quest Pro Independent Eye Gaze.

PCVR APPS NEEDED FOR THE BASELINE

Virtual Desktop for the Quest Pro

Meta Quest Store app on the headset; this is a paid app.

Virtual Desktop Streamer for Windows - direct EXE

Official Windows installer from the Virtual Desktop developer.

VRFaceTracking for Windows

Install through Steam, then add the Virtual Desktop module inside VRCFT.

SteamVR

Install through Steam on the Windows PC.

ONLY IF YOU NEED THESE

Singularity releases

For checking or updating your already installed Singularity APK.

Google Android Platform-Tools for Windows

For the manual ADB build check. Qpro's release already bundles the ADB files it needs.

CONNECTION REMINDER

Qpro-Enhanced-FT's enhanced eye and tongue tracking currently requires a **USB data cable** throughout the session. To stream Virtual Desktop PCVR over that cable too, switch the headset app to its **Beta** release channel and update it before starting. See page 5.

Before you install

ALREADY ROOTED

This guide starts after Singularity has rooted the **Quest Pro** and Singularity-Magisk appears under Unknown Sources. The enhanced app is Quest Pro-specific; do not apply these instructions to other Quest models. [1, 2]

WHAT YOU NEED

- Windows 10/11 x64 PC; SteamVR, Virtual Desktop on the headset plus its PC Streamer, and VRCFaceTracking (VRCFT). [2, 4, 5]
- Quest Pro eye and face tracking enabled; a **USB data cable**; Developer Mode and USB debugging authorization on this PC. [2, 4]
- Singularity-Magisk root access; later, grant **Shell / ADB Shell** Superuser access when QproFaceTracking requests it. [1, 2]

CHECK YOUR HEADSET FIRST

1

Turn on built-in tracking

In the Quest Pro Settings, open Movement Settings. Enable **Natural Facial Expressions** and **Eye Tracking**; run eye calibration if offered. [4]

2

Confirm normal PC tracking

In Virtual Desktop on the headset, go to **Streaming** and enable **Forward tracking data to PC**. Start the PC Streamer; install the **Virtual Desktop** module from VRCFT's Module Registry. Confirm ordinary facial tracking works before adding the enhanced app. For USB PCVR, use the Beta steps on page 5. [5]

3

Record your build

With USB debugging authorized, run `adb shell getprop ro.build.version.incremental` from your existing Platform-Tools folder. Singularity lists Quest Pro build 51483620027600340 as confirmed; the enhanced app's eye convergence was tested only on that same build. [1, 2]

FIRMWARE GATE

The Qpro README also says **v2.7 or lower** without identifying that version label in its installation steps. Do not equate it with the long build number. If your firmware differs from the tested build, treat eye convergence as unverified and check the maintainer's latest notes before trying it. [2]

WORKING PC EXAMPLE (OWNER REPORT)

OS: Windows 11 Pro 64-bit, version 25H2 (build 26200.9168)

CPU: Intel Core i7-13700KF (16 cores / 24 logical processors)

GPU: AMD Radeon RX 7900 XTX (driver 32.0.31041.3013)

Memory: 32 GB RAM; Gigabyte Z790 GAMING X AX motherboard

Owner's experience: Qpro-Enhanced-FT ran very well on this setup with very few issues. This is a personal result, not a measured performance benchmark.

Independent Eye Gaze

WHAT IT CHANGES

Singularity's Magisk Modules list points to Walaryne's **Quest Pro Independent Eye Gaze** module. The release says it stops averaging eye weights inside the internal Bolt model and disables a second averaging filter in trackingservice. This is intended to preserve separate left/right eye movement. [3, 6]

INSTALL IN THIS ORDER

1

Open Singularity's Magisk repo

On the rooted Quest Pro, open **Singularity > Apps/Modules > Magisk repo**. Both modules are listed there; this path does not require downloading ZIPs on a PC. [3]

2

Install Magisk OverlayFS first

Select **Magisk OverlayFS** and complete Singularity's install prompts. Reboot the headset, then check in **Singularity-Magisk > Modules** that OverlayFS is enabled before continuing. [3]

3

Install Independent Eye Gaze

Return to **Singularity > Apps/Modules > Magisk repo**. Select **Quest Pro Independent Eye Gaze** and complete the install. Reboot once more and confirm both modules are enabled in Singularity-Magisk. [3, 6]

4

Check eye behavior

Test both eyes in your normal VRCFT setup. If nearly closed eyes behave oddly, the module author recommends VRCFT filtering or local smoothing. [6]

HOW IT RELATES TO THE QPRO APP

- The Magisk module is a headset-level eye-processing change. Qpro-Enhanced-FT has its **own** gaze preparation and PC bridge; its README does not require this module. [2, 6]
- Their combined behavior is not documented. If Qpro gaze acts wrong, disable only Independent Eye Gaze in Magisk, reboot, and retest Qpro. Keep OverlayFS enabled for this setup. [2, 3, 6]

RECOVERY

If the module alone worsens eye behavior, disable Independent Eye Gaze in Singularity-Magisk, reboot, and retest normal tracking. Singularity's eye-calibration launcher is a separate feature.

Install Qpro-Enhanced-FT

USB-ONLY SOFTWARE

The release is an experimental **USB-connected** proof of concept for VRCFaceTracking. It can retain Virtual Desktop's normal face, brow, jaw, and blink data while optionally replacing left/right gaze and detailed tongue channels. The PC tool bundles Android Platform-Tools, so it does not need a second ADB install. [2]

ONE-TIME INSTALLATION

1

Download the complete release

Open the project's Releases page and get QproFaceTracking-<version>-poc.zip. Do **not** use GitHub's automatic Source code ZIP; it lacks the executable, model, Python installer, and Android tools. Extract the entire release folder. [2]

2

Set up the PC runtime

Run QproFaceTracking.exe from the extracted folder. Click **Set up PC runtime** and let it finish. The tool installs a private Python runtime and dependencies; the PyTorch download is large. No separate Python install or PATH edit is needed. [2]

3

Install the VRCFT bridge

Close VRCFaceTracking completely, click **Install/update bridge** in QproFaceTracking, then restart VRCFT. [2]

4

Prepare gaze only if using it

Keep the rooted headset connected by USB. Click **Prepare gaze from headset**. The tool reads your headset's stock eye archive and makes a local patch. Skip this step if you want only enhanced tongue tracking. [2]

ROOT PROMPT

When Singularity-Magisk asks whether **Shell / ADB Shell** may use Superuser, grant it for this setup. If the prompt never appears, inspect Singularity-Magisk's Superuser tab before retrying. [2]

ENABLE VIRTUAL DESKTOP BETA FOR USB PCVR

1

Switch the headset app to Beta

In the headset's app library, long-press **Virtual Desktop** and choose **Settings > Release Channels > Beta**. Alternatively, open its listing in the Meta Horizon phone app, expand **More Information**, tap the version, and select **Beta**. Then update the headset app. [11]

2

Update the PC Streamer and connect

Install the current Virtual Desktop Streamer from page 2. Launch the updated Beta app on the headset, connect the USB data cable, approve any USB connection prompt, and select the PC's USB connection if shown. This is the path the headset owner used for wired PCVR.

Run a tracking session

USB CARRIES THIS SETUP

In the owner's setup, Virtual Desktop **Beta** streamed the PCVR picture over USB while Qpro-Enhanced-FT used the live USB connection for enhanced eye/tongue data. Keep a USB **data** cable attached; wireless ADB is not a substitute for Qpro in this release. [2]

START IN THIS ORDER

1

Connect and authorize USB

Plug the Quest Pro into the Windows PC. Put on the headset and accept the USB debugging authorization for this PC if asked. Keep the cable attached for the entire session. [2]

2

Start Virtual Desktop over USB

Start Virtual Desktop Streamer on the PC. Open the updated Virtual Desktop Beta on the headset and choose its USB connection to your PC. Then start SteamVR and VRCFT. Verify ordinary eye/face tracking before enabling the enhanced features. [2, 5]

3

Choose enhancements

In QproFaceTracking, select **gaze**, **tongue**, or both, choose profiles/settings, and press **Apply and start selected**. Test eye movement and tongue movement in a VRCFT-compatible avatar. [2]

4

Stop cleanly

Before unplugging USB or closing the app, press **Stop and restore stock**. Then end VRCFT, SteamVR, and Virtual Desktop as usual. [2]

EXPECT TO PERSONALIZE THE DEMO DATA

- The included eye profile is calibrated for the developer, so gaze alignment may be off on another face or headset fit. [2]
- The included tongue v8 model was trained on one person. Start with **Weighted camera + native** and the conservative 24 FPS cap. If false positives or blind spots remain, use **Quick Refinement**. [2]
- For basic tongue personalization, the headset owner used **Quick Refinement** and got a very decent result. In the app, keep Virtual Desktop, SteamVR and VRCFT running, follow guided capture, name/select that dataset, run quick refinement, then choose the new model and test it. [2]

PRIVACY

Inward camera images are sensitive. The project says live frames stay local and recordings are created during explicit calibration. Do not share captures, training files, generated eye archives, or someone else's model without permission. [2]

Checks and download links

WHEN SOMETHING DOES NOT WORK

- **Qpro app cannot see headset:** use a USB data cable, keep Developer Mode on, accept the USB debugging prompt, and confirm Shell / ADB Shell in Magisk Superuser. [2]
- **Normal facial tracking fails:** fix Virtual Desktop forwarding and its VRCFT module first. Start Virtual Desktop Streamer before VRCFT. [5]
- **Virtual Desktop stays on Wi-Fi:** confirm the headset app is on the **Beta** release channel and has updated, the Windows Streamer is current, and the headset accepted USB connection access. [11]
- **Gaze patch fails or eye convergence looks wrong:** compare your build to 51483620027600340; disable the optional Magisk eye module for a clean Qpro test. Modern firmware is unverified by the Qpro author. [2]
- **Eyes behave oddly when nearly closed:** Walaryne notes this for the Magisk module and suggests VRCFT filtering or local smoothing. [6]
- **Tongue output is unreliable:** the bundled model is person-specific; use the app's guided calibration or quick refinement. [2]

OFFICIAL PROJECT PAGES AND DOWNLOADS

[1] Singularity root app

<https://github.com/Lumince/singularity>

[2] Qpro-Enhanced-FT complete release

<https://github.com/n0tmast3r/Qpro-Enhanced-FT/releases>

[3] Singularity Magisk Modules list

<https://github.com/Lumince/singularity/wiki/Magisk-Modules>

[4] Quest Pro tracking settings (VRCFT)

<https://docs.vrcft.io/docs/hardware/vr/meta/quest-pro>

[5] Virtual Desktop setup (VRCFT)

<https://docs.vrcft.io/docs/hardware/vr/meta/quest-pro/virtual-desktop>

[6] Independent Eye Gaze module release

<https://github.com/Walaryne/qp-magisk/releases>

[7] Singularity releases / module-repo change

<https://github.com/Lumince/singularity/releases>

[8] Virtual Desktop PC Streamer

<https://www.vrdesktop.net/>

[9] VRCFaceTracking download guide

<https://docs.vrcft.io/docs/intro/getting-started>

[10] SteamVR download

<https://store.steampowered.com/app/250820/SteamVR/>

[11] Meta: switch app release channels

<https://developers.meta.com/horizon/resources/publish-switch-release-channels/>